DC bridge specification

API description

Revision	Date	Comments
1.00	February 2022	Initial revision
1.01	March 2022	Changed device picture
1.02	March 2022	Corrections after review
1.03	April 2022	Reworked for customer communication protocol usage
1.04	September 2022	Reworked for v1.7 API version
1.05	December 2023	Reworked for v1.9 API version and AWS migration
1.06	March 2026	Migration SEQUANS cloud

Table of content

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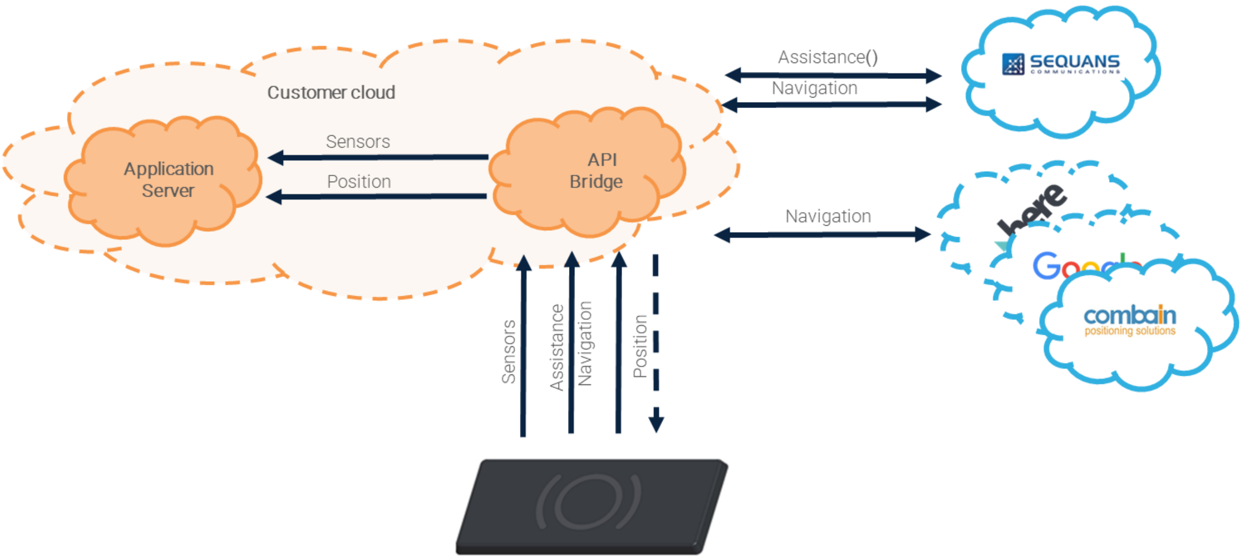
1 Introduction

This document is targeting ThinTrack customers who need to design a API bridge to interface between ThinTrack devices, Sequans GNSS Cloud, other location database (Combain, Google..) and their own server cloud.

Different messages are being exchanged between the devices, API bridge, devices and Sequans GNSS Cloud in detail:

- Devices to API bridge communication uses ThinTrack IP protocol (CoAP)
- API bridge communication to Sequans cloud for GNSS navigation (Https)
- API bridge communication to Combain database for WiFi/Cellular/Bluetooth navigation (Https)

The overall diagram is shown below:



The document provides a detailed description of the communication methods to be supported on that bridge, in order to communicate with devices and Sequans Cloud.

The DC bridge sources code are in https://github.com/Samea-Innovation/D2CBridge

2 ThinTrack IP communication protocol

2.1 Devices to API bridge communication

ThinTrack IP uses CoAP as the communication protocol between the devices and API bridge. API bridge must therefore embed a CoAP server configured as follows:

- CoAP port = 5683
- CoAP block size = 1024

The devices must be configured as follows:

- AT+LPGNSSCLOUDSEL=<bridge_hostname> where <bridge_hostname> is API bridge hostname (example: my.company.com)

API bridge must implement 3 CoAP APIs: assistance API and navigation API.

2.1.1 Assistance API

Resource	Value
URL	`coap://<bridge_hostname>:5683/vx.y/gnssAssistance`
Request type	POST
Payload	<assistance request payload to be forwarded to Sequans cloud assistance API, see 2.2.3>
Response	<assistance response payload received from Sequans cloud assistance API>

The assistance API is used to send a GNSS assistance request to Sequans cloud by using AT+LPGNSSASSISTANCE command on the device (see [1] for detailed syntax) and to get assistance data in return.

The assistance request payload received by the assistance API must be processed in API bridge and sent to Sequans cloud navigation API as explained in section 2.2.3.

The assistance response sent back by API bridge contains the assistance data to be used by ThinTrack IP.

2.1.2 Navigation API

Resource	Value
URL	`coap://<bridge_hostname>:5683/vx.y/gnssPosition`
Request type	POST
Payload	<navigation request payload to be forwarded to Sequans cloud navigation API, see 2.2.4>
Response	<navigation response payload received from Sequans cloud navigation API, if any. Empty otherwise>

The navigation API is used to send ThinTrack GNSS satellites measurements to Sequans cloud by using AT+LPGNSSSENDRAW or AT+LPGNSSCLOUDNAV commands on the device (see [1] for detailed syntax) and to get the device cloud-based position in return.

The navigation payload received by the navigation API must be processed in API bridge and sent to Sequans cloud navigation API as explained in section 2.2.4.

The navigation response sent by API bridge may contain a navigation payload (including the device position) that is used by ThinTrack IP. The device position is returned to the host application by the +LPGNSSSENDRAW or +LPGNSSCLOUDNAV URCs (see [] for detailed syntax).

2.1.1 Hybrid navigation API

Resource	Value
URL	`coap://<bridge_hostname>:5683/vx.y/locate`
Request type	POST
Payload	<hybrid navigation request payload to be forwarded to Sequans cloud hybrid navigation API, see 2.2.5>
Response	<hybrid navigation response payload received from Sequans cloud hybrid navigation API, if any. Empty otherwise>

The hybrid navigation API is used to send ThinTrack GNSS satellites measurements and optionally cellular and pressure information to Sequans cloud by using AT+LPGNSSSENDDATA command on the device (see [1] for detailed syntax) and to get the device cloud-based position in return.

The hybrid navigation payload received by the hybrid navigation API must be processed in API bridge and sent to Sequans cloud hybrid navigation API as explained in section 2.2.5.

The hybrid navigation response sent by API bridge may contain a hybrid navigation payload (including the device position) that is used by ThinTrack IP. The device position is returned to the device application by the +LPGNSSSENDATA URC (see [1] for detailed syntax).

2.2 API bridge to Sequans cloud communication

2.2.1 Authentication API

In order to communicate with Sequans cloud, API bridge must first proceed to an authentication phase by using the following authentication API:

Resource	Value
URL	`https://navigation.nw.do/authenticate`
Request type	POST

Body (raw type, json format)	`{`
	`  "username": "<user@company.com>",`
	`  "password": "<pwd>"`
	`}`
	where <user@company.com> and <pwd> are the credentials used during Sequans cloud account creation on https://gpsui.nw.do/

Response	`{`
	`  "token": "<access_token>"`
	`}`
	where <access_token> is the authorization token (valid for 30 days) to be used for further communication with Sequans cloud

2.2.1 Token renewal API

When expired, API bridge must renew the authorization token by using the following token renewal API:

Resource	Value
URL	`https://navigation.nw.do/authenticate/renew`
Request type	`POST`
Header key=Authorization	`Bearer <old_token>` where <old_token> is the old authorization token previously received in the authentication phase

Response	`{`
	`"token": "<new_token>"`
	`}`
	where <new_token> is the new authorization token (valid for 30 days) to be used for further communication with Sequans cloud

2.2.2 Assistance API

The assistance message payload received by API bridge CoAP server from the device must be processed as follows: remove 8 heading bytes and remove 4 tailing bytes.

The resulting bytes must be forwarded to Sequans cloud by using the following assistance API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/gnssAssistance` where vx.y is the API version received by API bridge CoAP server in the assistance API URL
Request type	`POST`
Header key=Authorization Content-Type=application/json	`Bearer <access_token>` where <access_token> is the authorization token received in the authentication phase

Body (json type)	`{`
	`"deviceId": "<unique device ID>",`
	`"assistancePayload": "<assistance request payload>"`
	`}`
	where:
	<unique device ID> is a unique ID identifying the device
	<assistance request payload> is the payload received by API bridge CoAP server on assistance API minus 8 heading bytes, minus 4 tailing bytes, then encoded in base64 format

Response	`<assistance response payload in binary format to be forwarded as is by API bridge CoAP server to the device in response to the CoAP assistance API>`

2.2.3 Navigation API

The navigation message payload received by API bridge CoAP server from the device must be processed as follows: remove 8 heading bytes and remove 4 tailing bytes.

The resulting bytes must be forwarded to Sequans cloud by calling the following navigation API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/gnssPosition` where vx.y is the API version received by API bridge CoAP server in the navigation API URL
Request type	`POST`
Header key=Authorization Content-Type=application/json	`Bearer <access_token>` where <access_token> is the authorization token received in the authentication phase

Body (json type)	`{`
	`"deviceId": "<device unique ID>",`
	`"rawMeas": "<navigation request payload>", "features": [<feature list>],`
	`"approxPos": [<longitude, latitude, altitude>]`
	`}`
	where:
	<unique device ID> is a unique ID identifying the device
	<navigation request payload> is the payload received by API bridge CoAP server on navigation API minus 8 heading bytes, minus 4 tailing bytes, then encoded in base64 format
	<feature list> is an optional coma-separated list of features to be activated on Sequans cloud (example: “NoC”, “AAN”)
	<longitude, latitude, altitude> is the optional device approximate position

Response	`<navigation response payload in json format containing the device position information, to be processed by API bridge (example: map display, log in a database…)>` NB: If the device position is requested by the device in the navigation request payload, then the navigation response payload contains an additional optional “payload” field. This payload must be forwarded as is by API bridge CoAP server to the device in CoAP navigation API response>.

Example of navigation API response payload:


```
{
    "utcTime":                          "2023-11-
    07T14:29:18.150949001+00:00",      "gpsTime":
    1383402576.150949,
    "confidence":11.193524,
    "position":[2.348045,48.853610,27.0836
    87],                     "velocity":[-
    0.095578,0.031254,0.303247],
    "gps":{
       "prn":[29,31,25,18,2,26,22],
       "cn0":[44,40,42,38,40,31,33]
    },
    "payload":
"MMEflqNKSEDcd5yPGlAEwNkq/kI/vsO9wgMAPSRDmz6tGDNBAQcdMGdPXQEdPgEfTgEZLQESR
A    EC9wAaCwEW",
    "HeightAboveTerrain":  22.439892425075655
```


Optional "payload" parameter to be forwarded to the device, if present.

2.2.1 Hybrid navigation API

The hybrid navigation message payload received by API bridge CoAP server from the device must be processed as follows: remove 8 heading bytes and remove 4 tailing bytes.

The resulting bytes must then be parsed and forwarded to Sequans cloud by calling the following hybrid navigation API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/locate` where vx.y is the API version received by API bridge CoAP server in the hybrid navigation API URL
Request type	`POST`
Header key=Authorization Content-Type=application/json	Bearer <access_token> where <access_token> is the authorization token received in the authentication phase

Body (json type)

```
{
 "deviceId": "<device unique ID>",
 "rawMeas": "<GNSS navigation
 payload>", "features": [<feature
 list>],
 "approxPos": [<longitude, latitude, altitude>],
 "hybrid": {
    "radioType": "LTE",
    "cellTowers": [{
       "cellId": <cell identifier>,
       "lac": <location area code>,
       "mcc": <mobile country code>,
       "mnc": <mobile network code>,
       "rsrp": <received power> }],
    "wifiAccessPoints": [{
       "mac": "<MAC address>",
       "rssi": <received power> }],
    "bluetoothBeacons": [{
       "mac": "<MAC address>",
       "rssi": <received power> }],
    "pressure": {
       "average": <average
       pressure>, "variance":
       <variance>, "count":
       <count>,
       "min": <minimum pressure>,
       "max": <maximum pressure> }}
}
```

where:
<unique device ID> is a unique ID identifying the device
<GNSS navigation payload> is the GNSS navigation payload received by API bridge CoAP server on hybrid navigation API encoded in base64 format
<feature list> is an optional coma-separated list of features to be activated on Sequans cloud (“NoC”, “AAN”, “PAAN”)
<longitude, latitude, altitude> is the optional device approximate position
<cell identifier> is the 28-bit LTE cell identifier
<location area code> is the LTE location area code
<mobile country code> is the LTE mobile country code
<mobile network code> is the LTE mobile network code
<received power> is the LTE or Wi-Fi or Bluetooth received power in dBm
<MAC address> is the Wi-Fi access point or Bluetooth beacon MAC address
<average pressure> is the average pressure in Pascal unit
<variance> is the pressure variance
<count> is the number samples used for average pressure computation
<minimum pressure> is the minimum pressure value
<maximum pressure> is the maximum pressure value

Response

`<hybrid navigation response payload in json format containing the device position information, to be processed by API bridge (example: map display, log in a database…)>`

NB: If the device position is requested by the device in the hybrid navigation request payload, then the hybrid navigation response payload contains an additional optional “payload” field. This payload must be forwarded as is by API bridge CoAP server to the device in CoAP hybrid navigation API response>.

Example of hybrid navigation API response payload:


```
{
    "utcTime":                              "2023-11-
    07T14:29:18.150949001+00:00",          "gpsTime":
    1383402576.150949,

    "confidence":11.193524,
    "position":[2.348045,48.853610,27.0836
    87],                        "velocity":[-
    0.095578,0.031254,0.303247],

    "gps":{

       "prn":[29,31,25,18,2,26,22],

       "cn0":[44,40,42,38,40,31,33]

    },

    "payload":
"MMEflqNKSEDcd5yPGlAEwNkq/kI/vsO9wgMAPSRDmz6tGDNBAQcdMGdPXQEdPgEfTgEZLQESR
A    EC9wAaCwEW",

    "HeightAboveTerrain":        22.439892425075655,
    “technology”: “GNSS”
```


Optional “payload” parameter to be forwarded to the device, if present.

3 Customer communication protocol

Customers may want to use their existing communication protocol between their devices and their cloud to piggyback GNSS data inside their existing data transfer packets.

3.1 Devices to customer cloud data

3 kinds of GNSS data can be piggybacked inside customer data transfer packets: assistance data, navigation data or hybrid navigation data. Together with any GNSS data, the API version information must also be piggybacked.

3.1.1 API version

The API version information to be piggybacked with any GNSS data is obtained using AT+LPGNSSCLOUDSEL? command on the device (see [1] for detailed syntax).

3.1.2 Assistance data

ThinTrack assistance request payload to be piggybacked inside uplink packets from the devices to customer cloud is obtained using AT+LPGNSSASSISTANCEPAYLOAD command on the device (see [1] for detailed syntax).

The assistance response payload to be piggybacked into devices downlink packets from customer cloud to devices is the data received from Sequans cloud in assistance API response (see section 3.2.2 for more details).

This assistance payload is then passed to ThinTrack IP using AT+LPGNSSSTOREASSISTANCE command on the device (see [1] for detailed syntax).

3.1.3 Navigation data

ThinTrack navigation request payload to be piggybacked into uplink packets from the devices to customer cloud is obtained by taking the <raw_meas> parameter in the response to AT+LPGNSSGETFIX command on the device (see [1] for detailed syntax).

The navigation response data to be piggybacked in devices downlink packets from customer cloud to devices depends on device use case:

- If the device position is not needed by the device, then there is nothing to piggyback to the device
- If the device position is needed by the device, then the customer cloud needs to:
    - Determine the validity of the position information sent by Sequans cloud:
        - -1000 < altitude < 10000
        - -138.9 < velocity < 138.9
        - Confidence < 10000
    - If the position information is valid then:
        - Send the position information such as latitude, longitude, altitude, velocity… (received from Sequans cloud in navigation API response) to the device application using customer communication protocol

Update ThinTrack IP approximate position of the device with latitude, longitude and altitude received from Sequans cloud in navigation API response by using AT+LPGNSSAPPROXPOS command on the device (see [1] for detailed syntax)

- If the position information is invalid, then it is up to customer to notify their device of the invalid position or not

3.1.4 Hybrid navigation data

ThinTrack hybrid navigation request payload to be piggybacked into uplink packets from the devices to customer cloud is obtained by taking the <raw_meas> parameter in the response to AT+LPGNSSGETFIX command on the device (see [1] for detailed syntax). More location data such as cellular, Wi-Fi, Bluetooth information or atmospheric pressure can also be added to the hybrid navigation payload.

The navigation response data to be piggybacked in devices downlink packets from customer cloud to devices depends on device use case:

- If the device position is not needed by the device, then there is nothing to piggyback to the device
- If the device position is needed by the device, then the customer cloud needs to:
  - Determine the validity of the position information sent by Sequans cloud:
    - -1000 < altitude < 10000
    - -138.9 < velocity < 138.9
    - Confidence < 10000
  - If the position information is valid then:
    - Send the position information such as latitude, longitude, altitude, velocity… (received from Sequans cloud in navigation API response) to the device application using customer communication protocol
    - Update ThinTrack IP approximate position of the device with latitude, longitude and altitude received from Sequans cloud in navigation API response by using AT+LPGNSSAPPROXPOS command on the device (see [1] for detailed syntax)
  - If the position information is invalid, then it is up to customer to notify their device of the invalid position or not

3.2 Customer cloud to Sequans cloud communication

3.2.1 Authentication API

In order to communicate with Sequans cloud, customer cloud must first proceed to an authentication phase with Sequans cloud by using the authentication API described in section 2.2.1.

3.2.2 Token renewal API

When expired, API bridge must renew the authorization token by using the token renewal API described in section 2.2.2.

3.2.3 Assistance API

GNSS assistance data piggybacked from the devices to customer cloud must be forwarded as is to Sequans cloud by using the following assistance API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/gnssAssistance` where vx.y is the API version piggybacked from device to customer cloud (see section 3.1.1)
Request type	`POST`
Header key=Authorizati on Content-Type=application /j son	`Bearer <access_token>` where <access_token> is the authorization token received in the authentication phase

Body (json type)	`{`
	`"deviceId": "<device unique ID>",`
	`"assistancePayload": "<assistance data>"`
	`}`
	where:
	<unique device ID> is a unique ID identifying the device
	<assistance data> is the assistance data piggybacked from device to customer cloud, see section 3.1.2

Response	`<assistance response payload in binary format to be forwarded as is by customer cloud to the device in response to the uplink piggybacked GNSS assistance request data>` see section 3.1.2 for more details

3.2.1 Navigation API

Any GNSS navigation data piggybacked from the devices to customer cloud must be forwarded as is to Sequans cloud by using the following navigation API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/gnssPosition` where vx.y is the API version piggybacked from device to customer cloud (see section 3.1.1)
Request type	POST
Header key=Authorization Content-Type=application/json	`Bearer <access_token>` where <access_token> is the authorization token received in the authentication phase
Body (json type)	`{ "deviceId": "<device unique ID>", "rawMeas": "<navigation data>", "features": [<feature list>], "approxPos": [<longitude, latitude, altitude>] }` where: <unique device ID> is a unique ID identifying the device <navigation data> is the navigation data piggybacked from device to customer cloud, see section 3.1.3 <feature list> is an optional coma-separated list of features to be activated on Sequans cloud (example: "NoC", "AAN") <longitude, latitude, altitude> is the optional device approximate position
Response	<navigation response payload in json format containing the device position information, to be processed by customer cloud (example: map display, log in a database, forward to device…) > see section 3.1.3 for more details


```
{
    "utcTime": "2023-11-07T14:29:18.150949001+00:00", "gpsTime": 1383402576.150949,
    "confidence":11.193524,
    "position": [2.348045,48.853610,27.083687],
    "velocity": [-0.095578,0.031254,0.303247],
    "gps":{
        "prn": [29,31,25,18,2,26,22],
        "cn0": [44,40,42,38,40,31,33]
    },
    "HeightAboveTerrain": 22.439892425075655
}
```

3.2.1 Hybrid navigation API

Any location data (GNSS, cellular, Wi-Fi, Bluetooth or atmospheric pressure) piggybacked from the devices to customer cloud must be forwarded to Sequans cloud by using the following navigation API:

Resource	Value
URL	`https://navigation.nw.do/vx.y/gnssPosition` where vx.y is the API version piggybacked from device to customer cloud (see section 3.1.1)
Request type	`POST`
Header key=Authorization Content-Type=application/json	`Bearer <access_token>` where <access_token> is the authorization token received in the authentication phase

Body (json type)	{
	“deviceId”: “<device unique ID>”,
	“rawMeas”: “<GNSS navigation payload>”, “features”: [<feature list>],
	“approxPos”: [<longitude, latitude, altitude>],
	"hybrid": {
	"radioType": "LTE",
	"cellTowers": [{
	"cellId": <cell identifier>,
	"lac": <location area code>,
	"mcc": <mobile country code>,
	"mnc": <mobile network code>,
	"rsrp": <received power> }],
	"wifiAccessPoints": [{
	"mac": "<MAC address>",
	"rssi": <received power> }],
	"bluetoothBeacons": [{
	"mac": "<MAC address>",
	"rssi": <received power> }],
	"pressure": {
	"average": <average pressure>, "variance": <variance>, "count": <count>,
	"min": <minimum pressure>,
	"max": <maximum pressure> }}
	}

	where: <unique device ID> is a unique ID identifying the device <GNSS navigation payload> is the GNSS navigation payload received by Sequans cloud bridge CoAP server on hybrid navigation API encoded in base64 format <feature list> is an optional coma-separated list of features to be activated on Sequans cloud (“NoC”, “AAN”, “PAAN”) <longitude, latitude, altitude> is the optional device approximate position <cell identifier> is the 28-bit LTE cell identifier <location area code> is the LTE location area code <mobile country code> is the LTE mobile country code <mobile network code> is the LTE mobile network code <received power> is the LTE or Wi-Fi or Bluetooth received power in dBm <MAC address> is the Wi-Fi access point or Bluetooth beacon MAC address <average pressure> is the average pressure in Pascal unit <variance> is the pressure variance <count> is the number samples used for average pressure computation <minimum pressure> is the minimum pressure value <maximum pressure> is the maximum pressure value
Response	`<navigation response payload in json format containing the device position information, to be processed by customer cloud (example: map display, log in a database, forward to device…)>` see section 3.1.4 for more details

Example of hybrid navigation API response payload:


```
{
    "utcTime": "2023-11-07T14:29:18.150949001+00:00", "gpsTime": 1383402576.150949,
    "confidence":11.193524,
    "position": [2.348045,48.853610,27.083687],
    "velocity": [-0.095578,0.031254,0.303247],
    "gps":{
      "prn": [29,31,25,18,2,26,22],
      "cn0": [44,40,42,38,40,31,33]
    },
    "HeightAboveTerrain": 22.439892425075655, “technology”: “GNSS”
}
```

3.3 Overall sequence